

JOC Computational Project: Milestone 3 Report

Phase Objective: 3/5 — Linear Modeling & Statistical Verification

Status: 100% COMPLETED

1. Mathematical Insights & Statistical Performance

In Phase 3, a Multiple Linear Regression (MLR) model was successfully trained using the scaled independent descriptors. The model achieved a robust **R² score of 0.509**. To meet rigorous JOC guidelines for descriptor independence, the correlation between steric size (D_max) and electronic polarization (Avg EN) was evaluated, showing an orthogonal value of **-0.292**. This low cross-correlation proves our model isolates distinct physical dimensions rather than capturing artificial overlapping trends.

2. Linear Regression Output & Residual Analysis Table

Compound ID	Substituent	D_max (Å)	Avg EN	Exp pKa	Pred pKa	Residual
1a	Methyl	3.54	2.35	25.0	23.82	1.18
1b	tert-Butyl	4.62	2.31	25.5	23.95	1.55
1c	Trifluoromethyl	4.394	3.113	18.2	16.48	1.72
1d	Phenyl	7.394	2.34	23.2	23.04	0.16
1e	p-Nitrophenyl	8.787	2.466	19.8	21.54	-1.74
1f	p-Methoxyphenyl	9.72	2.383	24.1	22.11	1.99
1g	Cyano	4.962	2.578	16.5	21.36	-4.86